



UAI301 Object Oriented Programming

Lab Assignment 2

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Subject Code: UAI301

Subject Name: Object Oriented Programming

Question 1 Write a Python program to print numbers from 0 to 10 using a while loop.

```
In [1]: i = 0

while i <= 10:
    print(i)
    i += 1
```

```
0
,1
,2
,3
,4
,5
,6
,7
,8
,9
,10
```

Question 2 Write a Python program to print numbers from 0 to 9 using a for loop.

```
In [2]: for i in range(10):
        print(i)
```

```
0
,1
,2
,3
,4
,5
,6
,7
,8
,9
```

Question 3 Write a Python program to print even numbers between 0 and 20 using the range() function.

```
In [3]: for i in range(0, 21, 2):
        print(i)
```

```
0
,2
,4
,6
,8
,10
,12
,14
,16
,18
,20
```

Question 4 Write a Python program to print numbers from 0 to -9 using a for loop.

```
In [4]: for i in range(0, -10, -1):
        print(i)
```

```
0
,-1
,-2
,-3
,-4
,-5
,-6
,-7
,-8
,-9
```

Question 5 Write a Python program to compare two numbers entered by the user.

```
In [5]: a = int(input("Enter first number: "))
        b = int(input("Enter second number: "))

        if a > b:
```

```
    print("First number is greater")
elif a < b:
    print("Second number is greater")
else:
    print("Both numbers are equal")
```

Enter first number: 5
,Enter second number: 6
Second number is greater

Question 6 Write a Python program to check whether a number is odd or even.

```
In [6]: num = int(input("Enter a number: "))

if num % 2 == 0:
    print("Even")
else:
    print("Odd")
```

Enter a number: 4
Even

Question 7 Write a Python program to check whether a number is prime or not.

```
In [7]: num = int(input("Enter a number: "))

if num < 2:
    print("Not Prime")
else:
    prime = True

    for i in range(2, int(num**0.5) + 1):
        if num % i == 0:
            prime = False
            break

    if prime:
        print("Prime")
    else:
        print("Not Prime")
```

Enter a number: 13
Prime

Question 8 Write a Python program to find the sum of all numbers divisible by 7 and 9 from 1 to n, where n is entered by the user.

```
In [8]: n = int(input("Enter n: "))

total = 0
```

```
for i in range(1, n + 1):
    if i % 7 == 0 and i % 9 == 0:
        total += i

print("Sum =", total)
```

Enter n: 63
Sum = 63

Question 9 Write a Python program using a function to find the sum of all odd numbers from 1 to n, where n is entered by the user.

```
In [9]: def odd_sum(n):
        total = 0

        for i in range(1, n + 1):
            if i % 2 != 0:
                total += i

        return total

n = int(input("Enter n: "))
print(odd_sum(n))
```

Enter n: 34
289

Question 10 Write a Python program using a function to find the sum of all prime numbers from 1 to n, where n is entered by the user.

```
In [10... def is_prime(num):
        if num < 2:
            return False

        for i in range(2, int(num**0.5) + 1):
            if num % i == 0:
                return False

        return True

def prime_sum(n):
    total = 0

    for i in range(2, n + 1):
        if is_prime(i):
            total += i

    return total

n = int(input("Enter n: "))
print(prime_sum(n))
```

Enter n: 13
41

Question 11 Write a Python program to display the length, uppercase, and lowercase forms of a string.

```
In [11... text = input("Enter a string: ")

print("Length:", len(text))
print("Uppercase:", text.upper())
print("Lowercase:", text.lower())
```

Enter a string: AmRitANShU
Length: 10
,Uppercase: AMRITANSHU
,Lowercase: amritanshu

Question 12 Write a Python program to split a string using spaces and commas.

```
In [12... text = input("Enter a string: ")

text = text.replace(",", " ")

print(text.split())
```

Enter a string: I love python
['I', 'love', 'python']

Question 13 Write a Python program to check whether a string is a palindrome.

```
In [13... text = input("Enter a string: ")

if text == text[::-1]:
    print("Palindrome")
else:
    print("Not Palindrome")
```

Enter a string: Naman
Not Palindrome

Question 14 Write a Python program to generate random integers between 1 and 100.

```
In [1]: import random

num = random.randint(1, 100)

print("Random integer:", num)
```

Random integer: 49

Question 15 Write a Python program to generate a random password using letters and digits.

```
In [2]: import random
import string

characters = string.ascii_letters + string.digits

password = ''.join(random.choice(characters) for i in range(8))

print("Random password:", password)
```

Random password: nb4iS0mv

Question 16 Write a Python program to generate a five-digit OTP.

```
In [3]: import random

otp = random.randint(10000, 99999)

print("Your OTP is:", otp)
```

Your OTP is: 63189

Question 17 Write a Python program to handle division by zero using exception handling.

```
In [14... try:
    a = int(input("Enter numerator: "))
    b = int(input("Enter denominator: "))

    print(a / b)

except ZeroDivisionError:
    print("Cannot divide by zero")
```

Enter numerator: 4
,Enter denominator: 0
Cannot divide by zero

Question 18 Write a Python program to handle an "array index out of range" exception.

```
In [15... numbers = [10, 20, 30]

try:
    print(numbers[5])

except IndexError:
    print("Index out of range")
```

Index out of range

Question 19 Write a Python program to handle the "File Not Found" exception.

```
In [16... try:
    file = open("data.txt", "r")
    print(file.read())
    file.close()

except FileNotFoundError:
    print("File not found")
```

File not found

Question 20 Write a Python program to create a list and display its elements.

```
In [17... numbers = [10, 20, 30, 40]

print(numbers)
```

[10, 20, 30, 40]

Question 21 Write a Python program to add and delete elements from a list.

```
In [18... numbers = [10, 20, 30]

numbers.append(40)

print(numbers)

numbers.remove(20)

print(numbers)
```

[10, 20, 30, 40]
,[10, 30, 40]

Question 22 Write a Python program to find the sum and average of list elements.

```
In [19... numbers = [10, 20, 30, 40]

total = sum(numbers)

average = total / len(numbers)

print("Sum =", total)
print("Average =", average)
```

Sum = 100
,Average = 25.0

Question 23 Write a Python program to find the maximum, minimum, and

sort a list in ascending and descending order.

```
In [20... numbers = [5, 8, 1, 9, 3]

print("Maximum =", max(numbers))
print("Minimum =", min(numbers))

numbers.sort()

print("Ascending =", numbers)

numbers.sort(reverse=True)

print("Descending =", numbers)
```

```
Maximum = 9
,Minimum = 1
,Ascending = [1, 3, 5, 8, 9]
,Descending = [9, 8, 5, 3, 1]
```

Question 24 Write a Python program to create a dictionary and access its values.

```
In [21... student = {
    "Name": "Rahul",
    "Age": 20,
    "Marks": 95
}

print(student["Name"])
print(student["Marks"])
```

```
Rahul
,95
```

Question 25 Write a Python program to display all keys and values of a dictionary.

```
In [22... student = {
    "Name": "Rahul",
    "Age": 20,
    "Marks": 95
}

for key, value in student.items():
    print(key, ":", value)
```

```
Name : Rahul
,Age : 20
,Marks : 95
```

Question 26 Write a Python program to update, add, delete, and clear

elements in a dictionary.

```
In [23... student = {
    "Name": "Rahul",
    "Age": 20
}

student["Age"] = 21

student["City"] = "Delhi"

del student["Name"]

print(student)

student.clear()

print(student)

{'Age': 21, 'City': 'Delhi'}
, {}
```

Question 27 Write a Python program to create and traverse a tuple.

```
In [24... numbers = (10, 20, 30, 40)

for i in numbers:
    print(i)

10
,20
,30
,40
```

Question 28 Write a Python program to search for an element in a tuple.

```
In [25... numbers = (10, 20, 30, 40)

element = int(input("Enter element: "))

if element in numbers:
    print("Found")
else:
    print("Not Found")

Enter element: 30
Found
```

Question 29 Write a Python program to add and delete elements in a tuple by converting it to a list and back to a tuple.

```
In [26... numbers = (10, 20, 30)
```

```
temp = list(numbers)
temp.append(40)
temp.remove(20)
numbers = tuple(temp)
print(numbers)

(10, 30, 40)
```

Question 30 Write a Python program to create a set and perform union, intersection, difference, and symmetric difference operations.

```
In [27... A = {1, 2, 3, 4}
B = {3, 4, 5, 6}

print("Union:", A | B)
print("Intersection:", A & B)
print("Difference:", A - B)
print("Symmetric Difference:", A ^ B)

Union: {1, 2, 3, 4, 5, 6}
,Intersection: {3, 4}
,Difference: {1, 2}
,Symmetric Difference: {1, 2, 5, 6}
```

Question 31 Write a Python program to add, remove, and pop elements from a set.

```
In [28... numbers = {10, 20, 30}

numbers.add(40)

print(numbers)

numbers.remove(20)

print(numbers)

numbers.pop()

print(numbers)

{40, 10, 20, 30}
,{40, 10, 30}
,{10, 30}
```